Analysis of Linear Programming

Part 3e: Matrix Form, Dual Simplex and Revised Simplex

IE 311 – Operations Research I

Different Versions of the Simplex Method

- So far we have learned the (primal) simplex method as the only solution procedure for linear programs.
- We now discuss different versions of the simplex method:
- first we derive the **matrix form** of the (primal) simplex method;
- then we discuss the **optimality conditions** in linear programming.
- Finally, combining the above, we reach the **Dual Simplex** and **Revised Simplex** methods.

Matrix Form of the Simplex Tableau

- Let us consider an LP in standard form (assume that c is an n-row vector, b an is a m-column vector and A is an $m \times n$ matrix):

$$\begin{aligned} \max\ x_0 &= cx \\ \text{s.t.} \qquad Ax &= b \\ x &\geq 0. \end{aligned}$$

- Suppose that we are given a basic solution where the basic variables are indexed by the set BV.
- <u>Question</u>: How can we construct the simplex tableau via matrix operations?

Matrix Form of the Simplex Tableau

- Recall the LP in standard form:

$$\begin{aligned} \max\ x_0 &= cx \\ \text{s.t.} \quad Ax &= b \\ x &\geq 0. \end{aligned}$$

- Let us partition the variables as basic, denoted by x_{BV}, and nonbasic, denoted by x_{NBV} (also partition c and A accordingly).

$$\begin{aligned} \max\ x_0 &= c_{\text{BV}} x_{\text{BV}} + c_{\text{NBV}} x_{\text{NBV}} \\ \text{s.t.} \quad B x_{\text{BV}} + \quad N x_{\text{NBV}} &= b \\ x_{\text{BV}},\ x_{\text{NBV}} &\geq 0. \end{aligned}$$

- Since B is a basis matrix (B^{-1} exists), we have $x_{\text{BV}} = B^{-1}(b - N x_{\text{NBV}})$.

Matrix Form of the Simplex Tableau

- If we plug in $x_{\text{BV}} = B^{-1}(b - Nx_{\text{NBV}})$ into the following LP,

$$\begin{aligned} \max\ x_0 &= c_{\text{BV}}x_{\text{BV}} + c_{\text{NBV}}x_{\text{NBV}} \\ \text{s.t.} \quad & Bx_{\text{BV}} + \quad Nx_{\text{NBV}} = b \\ & x_{\text{BV}},\ x_{\text{NBV}} \geq 0. \end{aligned}$$

- we obtain,

$$\begin{aligned} \max\ x_0 \quad & + (c_{\text{BV}}B^{-1}N - c_{\text{NBV}})x_{\text{NBV}} = c_{\text{BV}}B^{-1}b \\ \text{s.t.} \quad x_{\text{BV}} + & \qquad (B^{-1}N)x_{\text{NBV}} = B^{-1}b \\ & x_{\text{BV}},\ x_{\text{NBV}} \geq 0. \end{aligned}$$

- or in tableau form:

Row	BV		RHS
0	x_0	$x_0 \quad +(c_{\text{BV}}B^{-1}N - c_{\text{NBV}})x_{\text{NBV}}$	$= c_{\text{BV}}B^{-1}b$
$1,\ldots,m$	x_{BV}	$x_{\text{BV}} \quad +(B^{-1}N)x_{\text{NBV}}$	$= B^{-1}b$

Matrix Form of the Simplex Tableau: Summary

- Given a basis matrix B (corresponding to the columns in BV), we obtain the following simplex tableau:

Row	BV		RHS
0	x_0	$x_0 \quad +(c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}})x_{\mathrm{NBV}}$	$= c_{\mathrm{BV}}B^{-1}b$
$1,\ldots,m$	x_{BV}	$x_{\mathrm{BV}} \quad +(B^{-1}N)x_{\mathrm{NBV}}$	$= B^{-1}b$

- If we have $B^{-1}b \geq 0$, then BV gives a **basic feasible solution**.
- The row vector $\bar{c} := c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}}$ is called *the reduced cost*.
- Reduced cost of variable j is computed as $\bar{c}_j := c_{\mathrm{BV}}B^{-1}A^j - c_j$, where A^j is the corresponding column of the matrix A.
- Note that the current tableau is **optimal if all reduced costs are nonnegative**.

Matrix Form of the Simplex Tableau: Example

- Consider the following LP in standard form:

$$\begin{aligned} \max\ x_0 &= 3x_1 - 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 2 \\ & x_1 + x_4 = 1 \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

- Construct the simplex tableau via matrix operations for
  - BV = {1,2}, NBV = {3,4}
  - BV = {2,1}, NBV = {3,4}
  - BV = {1,3}, NBV = {2,4}
  - BV = {2,4}, NBV = {1,3}.

Matrix Form of the Simplex Tableau: Example (continued)

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_{\mathrm{BV}} = \begin{bmatrix} 3 & -2 \end{bmatrix}, c_{\mathrm{NBV}} = \begin{bmatrix} 0 & 0 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, B^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix},$$

$$B^{-1}N = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} -2 & 5 \end{bmatrix}, B^{-1}b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = 1.$$

Row	BV						RHS
0	x_0	x_0			$-2x_3$	$+5x_4$	$= 1$
1	x_1		x_1			$+x_4$	$= 1$
2	x_2			x_2	$+x_3$	$-x_4$	$= 1$

Matrix Form of the Simplex Tableau: Example (continued)

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_{\mathrm{BV}} = \begin{bmatrix} -2 & 3 \end{bmatrix}, c_{\mathrm{NBV}} = \begin{bmatrix} 0 & 0 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, B^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix},$$

$$B^{-1}N = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} -2 & 5 \end{bmatrix}, B^{-1}b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = 1.$$

Row	BV					RHS
0	x_0	x_0		$-2x_3$	$+5x_4$	$= 1$
1	x_2		x_2	$+x_3$	$-x_4$	$= 1$
2	x_1	x_1			$+x_4$	$= 1$

Matrix Form of the Simplex Tableau: Example (continued)

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_{\mathrm{BV}} = \begin{bmatrix} 3 & 0 \end{bmatrix}, c_{\mathrm{NBV}} = \begin{bmatrix} -2 & 0 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, B^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix},$$

$$B^{-1}N = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} 2 & 3 \end{bmatrix}, B^{-1}b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = 3.$$

Row	BV						RHS
0	x_0	x_0		$+2x_2$		$+3x_4$	$= 3$
1	x_1		x_1			$+x_4$	$= 1$
2	x_3			x_2	$+x_3$	$-x_4$	$= 1$

Matrix Form of the Simplex Tableau: Example (continued)

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, N = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, c_{\mathrm{BV}} = \begin{bmatrix} -2 & 0 \end{bmatrix}, c_{\mathrm{NBV}} = \begin{bmatrix} 3 & 0 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, B^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$B^{-1}N = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} -5 & -2 \end{bmatrix}, B^{-1}b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = -4.$$

Row	BV						RHS
0	x_0	x_0	$-5x_1$		$-2x_3$		$= -4$
1	x_2		x_1	$+x_2$	$+x_3$		$= 2$
2	x_4		x_1			$+x_4$	$= 1$

Matrix Form of the Simplex Tableau: Example (continued)

Exercise: Try to construct the simplex tableau via matrix operations for the following selections. Comment on which ones lead to basic solutions, basic feasible solutions or none.

- $\mathrm{BV} = \{3,1\}, \mathrm{NBV} = \{2,4\}$
- $\mathrm{BV} = \{3,4\}, \mathrm{NBV} = \{1,2\}$
- $\mathrm{BV} = \{1,4\}, \mathrm{NBV} = \{2,3\}$
- $\mathrm{BV} = \{2,3\}, \mathrm{NBV} = \{1,3\}.$

Matrix Form of the Simplex Tableau: Relation with the Dual

- Consider the following primal-dual pair (here, y is a row-vector):

$$\begin{aligned} \max \quad & cx \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned} \qquad \begin{aligned} \min \quad & yb \\ \text{s.t.} \quad & yA \geq c \\ & y \text{ urs.} \end{aligned}$$

- Given BV, we can rewrite as follows:

$$\begin{aligned} \max \quad & c_{\text{BV}} x_{\text{BV}} + c_{\text{NBV}} x_{\text{NBV}} \\ \text{s.t.} \quad & B x_{\text{BV}} + N x_{\text{NBV}} = b \\ & x_{\text{BV}},\ x_{\text{NBV}} \geq 0. \end{aligned} \qquad \begin{aligned} \min \quad & yb \\ \text{s.t.} \quad & yB \geq c_{\text{BV}},\ yN \geq c_{\text{NBV}} \\ & y \text{ urs.} \end{aligned}$$

Matrix Form of the Simplex Tableau: Relation with the Dual

- Given BV, we can rewrite primal-dual pair as follows:

$$\begin{aligned} \max \quad & c_{\mathrm{BV}} x_{\mathrm{BV}} + c_{\mathrm{NBV}} x_{\mathrm{NBV}} \\ \text{s.t.} \quad & B x_{\mathrm{BV}} + N x_{\mathrm{NBV}} = b \\ & x_{\mathrm{BV}},\ x_{\mathrm{NBV}} \geq 0. \end{aligned} \qquad \begin{aligned} \min \ & yb \\ \text{s.t.} \ & yB \geq c_{\mathrm{BV}},\ yN \geq c_{\mathrm{NBV}} \\ & y \text{ urs.} \end{aligned}$$

- Let us *predict* a dual solution as $y_{\mathrm{BV}} := c_{\mathrm{BV}} B^{-1}$.
- When initialized with a BFS, the simplex method guarantees that
  - $(x_{\mathrm{BV}}, x_{\mathrm{NBV}}) := (B^{-1}b, 0)$ is feasible at every iteration
  - complementary slackness conditions hold between $(x_{\mathrm{BV}}, x_{\mathrm{NBV}})$ and y_{BV}
  - we terminate when the dual solution $y_{\mathrm{BV}} := c_{\mathrm{BV}} B^{-1}$ becomes feasible.

CLAIM: The simplex method guarantees that complementary slackness conditions $(yA - c)_j\, x_j = 0$ (or equivalently $\bar{c}_j x_j = 0$) hold between $(x_{\mathrm{BV}}, x_{\mathrm{NBV}}) := (B^{-1}b, 0)$ and $y_{\mathrm{BV}} := c_{\mathrm{BV}} B^{-1}$.

Row	BV			RHS
0	x_0	x_0	$+(c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}})x_{\mathrm{NBV}}$	$= c_{\mathrm{BV}}B^{-1}b$
$1,\ldots,m$	x_{BV}	x_{BV}	$+(B^{-1}N)x_{\mathrm{NBV}}$	$= B^{-1}b$

- **PROOF**:
  - If x_j is basic, then its reduced cost $\bar{c}_j$ is 0.
  - If x_j is nonbasic, then its value is 0.

Matrix Form of the Simplex Tableau: Dual Feasibility upon Optimality

- **CLAIM**: The simplex method terminates when the dual solution $y_{\text{BV}} := c_{\text{BV}}B^{-1}$ becomes feasible.

Row	BV			RHS
0	x_0	x_0	$+(c_{\text{BV}}B^{-1}N - c_{\text{NBV}})x_{\text{NBV}}$	$= c_{\text{BV}}B^{-1}b$
$1,\ldots,m$	x_{BV}	x_{BV}	$+(B^{-1}N)x_{\text{NBV}}$	$= B^{-1}b$

- **PROOF**: At termination, we have $\bar{c} := c_{\text{BV}}B^{-1}N - c_{\text{NBV}} \geq 0$. Since
  - $yB = c_{\text{BV}}B^{-1}B \geq c_{\text{BV}}$, and $yN = c_{\text{BV}}B^{-1}N \geq c_{\text{NBV}}$,

  $y_{\text{BV}} := c_{\text{BV}}B^{-1}$ is a feasible solution to the dual problem:

$$\begin{aligned} \min\ & yb \\ \text{s.t. } & yB \geq c_{\text{BV}},\ \ yN \geq c_{\text{NBV}} \\ & y \text{ urs.} \end{aligned}$$

- **CLAIM**: Strong duality *holds* between $(x_{\text{BV}}, x_{\text{NBV}}) := (B^{-1}b, 0)$ and $y_{\text{BV}} := c_{\text{BV}}B^{-1}$ at every iteration.

Row	BV			RHS
0	x_0	x_0	$+(c_{\text{BV}}B^{-1}N - c_{\text{NBV}})x_{\text{NBV}}$	$= c_{\text{BV}}B^{-1}b$
$1,\ldots,m$	x_{BV}	x_{BV}	$+(B^{-1}N)x_{\text{NBV}}$	$= B^{-1}b$

- **PROOF**:

$$cx = c_{\text{BV}}x_{\text{BV}} = c_{\text{BV}}\,(B^{-1}b) = (c_{\text{BV}}B^{-1})b = y_{\text{BV}}b = yb.$$

Matrix Form of the Simplex Tableau: A Special Case

- Consider the standard form of a well-structured LP (only $\leq$ constraints with nonnegative RHS):

$$\begin{aligned} \max\ x_0 &= cx \\ \text{s.t.} \qquad Ax + I\tilde{x} &= b \\ x \geq 0, \tilde{x} &\geq 0. \end{aligned}$$

- Then,
  - values of the dual variables are equal to the reduced cost of the slack variables since $\bar{c}_{\text{slack}} := c_{\text{BV}} B^{-1} I - c_{\text{slack}} = c_{\text{BV}} B^{-1} = y_{\text{BV}}$ (note $c_{\text{slack}} = 0$)
  - the basis inverse can be retrieved from the columns under the slack variables since $B^{-1} I = B^{-1}$.
- See the example above!

Optimality Conditions in Linear Programming

- Consider the following primal-dual pair (here, y is a row-vector):

$$\begin{aligned} \max \quad & cx \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned} \qquad\qquad \begin{aligned} \min \quad & yb \\ \text{s.t.} \quad & yA \geq c \\ & y \text{ urs.} \end{aligned}$$

- A primal-dual pair (x, y) is **optimal** if and only if
  1. x is <u>primal feasible</u>, that is, $Ax = b, x \geq 0$
  2. y is <u>dual feasible</u>, that is, $yA \geq c$
  3. (x, y) satisfies <u>complementary slackness</u>, that is, $(yA - c)_j \, x_j = 0$.
  - <u>Complementary slackness</u> is equivalent to <u>strong duality</u>, that is, $yb = cx$.

(Primal) Simplex Method vs. Dual Simplex Method

- **(Primal) Simplex Method:**
- keeps **primal feasibility** and complementary slackness intact at each iteration, and
- terminates when **dual feasibility** is achieved.
- There is an alternative method for LPs with appropriate structure, the **Dual Simplex Method**.
- **Dual Simplex Method:**
- keeps **dual feasibility** and complementary slackness intact at each iteration, and
- terminates when **primal feasibility** is achieved.

(Primal) Simplex Method: Reminder

The simplex algorithm proceeds as follows:

- **Step 1:** Convert the LP to standard form.
- **Step 2:** Obtain a BFS (if possible) from the standard form.
- **Step 3:** Determine whether the current BFS is optimal.
- **Step 4:** If not optimal, determine which nonbasic variable should become basic and which basic variable should become nonbasic, to obtain a better objective value (or detect unboundedness).
- **Step 5:** Use EROs to obtain that new BFS, then return to Step 3.

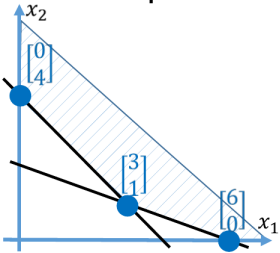
Dual Simplex Method

The dual simplex algorithm proceeds as follows:

- **Step 1:** Convert the LP to standard form.
- **Step 2:** Obtain a **dual BFS** from the standard form.
- **Step 3:** Determine whether the current dual BFS is **primal feasible**.
- **Step 4:** If not primal feasible, determine which basic variable should become nonbasic and which nonbasic variable should become basic, to obtain a new dual BFS with a worse objective value (or detect infeasibility).
- **Step 5:** Use EROs to obtain that new dual BFS, then return to Step 3.

- Solve the following problem using the dual simplex method:

$$\begin{aligned} \min\ x_0 = x_1 + \ \ x_2 \\ \text{s.t.} \qquad x_1 + \ \ x_2 \geq 4 \\ x_1 + 3x_2 \geq 6 \\ x_1, x_2 \geq 0. \end{aligned}$$



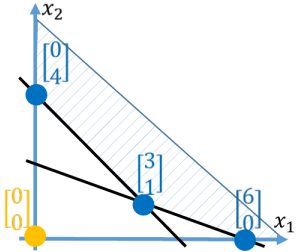
- In Step 1, we transform the problem into the following standard form of a maximization LP (note the negative RHS vector):

$$\begin{aligned} \max\ x_0 = -x_1 - \ \ x_2 \\ \text{s.t.} \qquad -x_1 - \ \ x_2 + x_3 \qquad = -4 \\ -x_1 - 3x_2 \qquad + x_4 = -6 \end{aligned}$$

Dual Simplex Example: Step 2

- In Step 2, we obtain an initial dual BFS:

Row	BV						RHS
0	x_0'	x_0'	$+x_1$	$+x_2$			$= \quad 0$
1	x_3		$-x_1$	$-x_2$	$+x_3$		$= -4$
2	x_4		$-x_1$	$-3x_2$		$+x_4$	$= -6$



- **Note**: The slack variables x_3 and x_4 are chosen as basic variables.
- **Reduced costs** of the slack variables correspond to the **dual variables** (in this iteration, the origin is a dual BFS).
- **RHS column** in this tableau is not nonnegative.

Dual Simplex Example: Step 3

- In Step 3, we check whether the current dual BFS is primal feasible (**look at RHS**):

Row	BV						RHS
0	x_0'	x_0'	$+x_1$	$+x_2$			$= 0$
1	x_3		$-x_1$	$-x_2$	$+x_3$		$= -4$
2	x_4		$-x_1$	$-3x_2$		$+x_4$	$= -6$

- **Idea**: If we move to an adjacent dual BFS, can we decrease the *infeasibility* (make the RHS less negative while keeping reduced costs nonnegative)?
- Moving to an adjacent BFS means that we make one of the basic variables nonbasic while making one of the nonbasic variables basic.
- **Note that the values of x_3 and x_4 are negative.**
- Therefore, the current dual BFS is NOT primal feasible.

- In Step 4, we first decide which basic variable to make nonbasic (**look at RHS**):

Row	BV						RHS
0	x_0'	x_0'	$+x_1$	$+x_2$			$= \quad 0$
1	x_3		$-x_1$	$-x_2$	$+x_3$		$= -4$
2	x_4		$-x_1$	$-3x_2$		$+x_4$	$= -6$

- **Idea**: Intuitively, choosing the most negative RHS might decrease the infeasibility the most (although any basic variable with negative RHS can be chosen).
- **In this case, we can choose the basic variable x_4 to leave the basis, hence, make it nonbasic.**

Dual Simplex Example: Step 4

- In Step 4, we next decide which nonbasic variable to make basic (**look at the row of the leaving variable and row-0**):

Row	BV						RHS
0	x_0'	x_0'	$+x_1$	$+x_2$			$= 0$
1	x_3		$-x_1$	$-x_2$	$+x_3$		$= -4$
2	x_4		$-x_1$	$-3x_2$		$+x_4$	$= -6$

- **Idea**: What is the adjacent BFS obtained by making x_4 nonbasic?
- We need to perform the so-called ***minimum ratio test***:
  - Obtain the negative of the entrywise ratio of row-0 and the row of the leaving variable among the columns that have negative entry in the row of the leaving variable.
  - Choose the nonbasic variable that gives the minimum ratio as the entering variable (if there is no such variable, dual is unbounded and primal is infeasible):

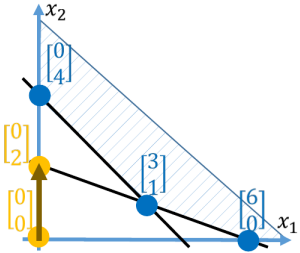
Dual Simplex Example: Step 5

- In Step 5, we use EROs to find a new BFS with a *worse* objective function value (**pivoting**):

Row	BV						RHS
0	x_0'	x_0'	$+x_1$	$+x_2$			$= 0$
1	x_3		$-x_1$	$-x_2$	$+x_3$		$= -4$
2	x_4		$-x_1$	$-3x_2$		$+x_4$	$= -6$

- **Idea**: How can we make the column of the entering variable a unit vector by pivoting?

Row	BV						RHS
0	x_0'	x_0'	$+\frac{2}{3}x_1$			$+\frac{1}{3}x_4$	$= -2$
1	x_3		$-\frac{2}{3}x_1$		$+x_3$	$-\frac{1}{3}x_4$	$= -2$
2	x_2		$\frac{1}{3}x_1$	$+x_2$		$-\frac{1}{3}x_4$	$= 2$



Dual Simplex Example: Step 3 (Again)

- In Step 3, we check whether the current BFS is primal feasible (**look at RHS**):

Row	BV						RHS
0	x_0'	x_0'	$+\frac{2}{3}x_1$			$+\frac{1}{3}x_4$	$= -2$
1	x_3		$-\frac{2}{3}x_1$		$+x_3$	$-\frac{1}{3}x_4$	$= -2$
2	x_2		$\frac{1}{3}x_1$	$+x_2$		$-\frac{1}{3}x_4$	$= 2$

- **Idea**: If we move to an adjacent dual BFS, can we decrease the *infeasibility* (make the RHS less negative while keeping reduced costs nonnegative)?
- **Note that the value of x_3 is negative.**
- Therefore, the current dual BFS is NOT primal feasible.

Dual Simplex Example: Step 4 (Again)

- In Step 4, we first decide which basic variable to make nonbasic (**look at RHS**):

Row	BV						RHS
0	x_0'	x_0'	$+\frac{2}{3}x_1$			$+\frac{1}{3}x_4$	$= -2$
1	x_3		$-\frac{2}{3}x_1$		$+x_3$	$-\frac{1}{3}x_4$	$= -2$
2	x_2		$\frac{1}{3}x_1$	$+x_2$		$-\frac{1}{3}x_4$	$= 2$

- **Idea**: Intuitively, choosing the most negative RHS might decrease the infeasibility the most (although any basic variable with negative RHS can be chosen).
- **In this case, there is only one candidate: the basic variable x_3 leaves the basis.**

Dual Simplex Example: Step 4 (Again)

- In Step 4, we next decide which nonbasic variable to make basic (**look at the row of the leaving variable and row-0**):

Row	BV						RHS
0	x_0'	x_0'	$+\frac{2}{3}x_1$			$+\frac{1}{3}x_4$	$= -2$
1	x_3		$-\frac{2}{3}x_1$		$+x_3$	$-\frac{1}{3}x_4$	$= -2$
2	x_2		$\frac{1}{3}x_1$	$+x_2$		$-\frac{1}{3}x_4$	$= \ \ 2$

- **Idea**: What is the adjacent BFS obtained by making x_3 nonbasic?
- We perform the so-called ***minimum ratio test***:

$$\text{There is a tie since } \min\left\{-\frac{2/3}{-2/3}, -\frac{1/3}{-1/3}\right\} = 1.$$

- We choose $\boldsymbol{x_1}$ to enter the basis (we could have also chosen $\boldsymbol{x_4}$).
- Note: The next dual BFS will be degenerate

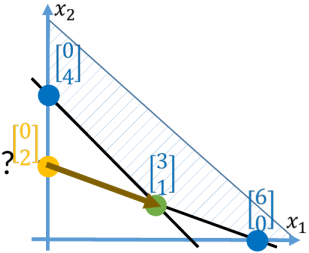
Dual Simplex Example: Step 5 (Again)

- In Step 5, we use EROs to find a new BFS with a *worse* objective function value (**pivoting**):

Row	BV						RHS
0	x_0'	x_0'	$+\frac{2}{3}x_1$			$+\frac{1}{3}x_4$	$= -2$
1	x_3		$-\frac{2}{3}x_1$		$+x_3$	$-\frac{1}{3}x_4$	$= -2$
2	x_2		$\frac{1}{3}x_1$	$+x_2$		$-\frac{1}{3}x_4$	$= 2$

- **Idea**: How can we make the column of the entering variable a unit vector by pivoting?

Row	BV						RHS
0	x_0'	x_0'			$+x_3$		$= -4$
1	x_1		x_1		$-\frac{3}{2}x_3$	$+\frac{1}{2}x_4$	$= 3$
2	x_2			$+x_2$	$+\frac{1}{2}x_3$	$-\frac{1}{2}x_4$	$= 1$



Dual Simplex Example: Step 3 (Again)

- In Step 3, we check whether the current BFS is primal feasible (**look at RHS**):

Row	BV						RHS
0	x_0'	x_0'			$+x_3$		$= -4$
1	x_1		x_1		$-\frac{3}{2}x_3$	$+\frac{1}{2}x_4$	$= 3$
2	x_2			$+x_2$	$+\frac{1}{2}x_3$	$-\frac{1}{2}x_4$	$= 1$

- **Note that the values of x_1 and x_2 are nonnegative, hence, we obtain a primal feasible solution.**
- Therefore, the current BFS is **OPTIMAL** with objective value 4.
- Optimal primal solution: $x_1 = 3,\ x_2 = 1$.

Recall: Matrix Form of the Simplex Tableau

- Consider the following LP:

$$\begin{aligned} \max\ x_0 &= cx \\ \text{s.t.} \quad Ax &= b \\ x &\geq 0. \end{aligned}$$

- Given a basis BV, we can rewrite the LP as follows:

$$\begin{aligned} \max\ x_0 \quad &+ (c_{\mathrm{BV}} B^{-1} N - c_{\mathrm{NBV}}) x_{\mathrm{NBV}} = c_{\mathrm{BV}} B^{-1} b \\ \text{s.t.} \quad x_{\mathrm{BV}} + \quad &(B^{-1} N) x_{\mathrm{NBV}} = B^{-1} b \\ x_{\mathrm{BV}},\ x_{\mathrm{NBV}} &\geq 0. \end{aligned}$$

- We will now see a version of the simplex method (called the *Revised Simplex Method*) which does NOT use tableaus at all.

Revised Simplex Method

The revised simplex algorithm proceeds as follows:

- **Step 1** Convert the LP to standard form.
- **Step 2** Obtain an initial feasible basis BV.
- **Step 3** Given BV, compute basis matrix inverse B^{-1}, primal solution $B^{-1}b$, reduced cost $\bar{c} := c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}}$ and objective value $c_{\mathrm{BV}}B^{-1}b$.
  Determine whether the current BFS is optimal (check if there exists $\bar{c}_j < 0$).
- **Step 4** If the current BFS is not optimal, then determine which nonbasic variable (say j^E) should become a basic variable.
  Compute the column of the entering variable $B^{-1}A^{j^E}$.
  Use minimum ratio test to decide which basic variable (say j^L) should become a nonbasic variable (or determine if the problem is unbounded).
- **Step 5** Update $\mathrm{BV} = \mathrm{BV} \cup \{j^E\} \setminus \{j^L\}$. Go back to step 3.

- Solve the following problem using the revised simplex method:

$$\begin{aligned} \max\ x_0 &= 5x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 6x_2 \leq 12 \\ & x_1, x_2 \geq 0. \end{aligned}$$

- In Step 1, we transform the problem into the following standard form:

$$\begin{aligned} \max\ x_0 &= 5x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 + x_3 = 12 \\ & x_1 + 6x_2 + x_4 = 12 \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

Revised Simplex Method: Step 2

- Here is the matrix form of the LP:

$$A = \begin{bmatrix} 3 & 2 & 1 & 0 \\ 1 & 6 & 0 & 1 \end{bmatrix}, c = \begin{bmatrix} 5 & 4 & 0 & 0 \end{bmatrix}, b = \begin{bmatrix} 12 \\ 12 \end{bmatrix}.$$

- In Step 2, we should obtain an initial feasible BFS.
- **Note**: The slack variables x_3 and x_4 can be chosen as basic variables since the RHS is nonnegative.
- Hence, $\mathrm{BV} = \{3,4\}$.

Revised Simplex Method: Iteration 1 (Step 3)

- Recall the matrix form:

$$A = \begin{bmatrix} 3 & 2 & 1 & 0 \\ 1 & 6 & 0 & 1 \end{bmatrix}, c = \begin{bmatrix} 5 & 4 & 0 & 0 \end{bmatrix}, b = \begin{bmatrix} 12 \\ 12 \end{bmatrix}.$$

- We compute the following matrices for BV = {3,4}:

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, N = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}, c_{\mathrm{BV}} = \begin{bmatrix} 0 & 0 \end{bmatrix}, c_{\mathrm{NBV}} = \begin{bmatrix} 5 & 4 \end{bmatrix}, B^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$B^{-1}N = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}, B^{-1}b = \begin{bmatrix} 12 \\ 12 \end{bmatrix}, \bar{c} = c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} -5 & -4 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = 0.$$

- Reduced cost $\bar{c}$ is NOT nonnegative. Current BV is not proven to be optimal.

Revised Simplex Method: Iteration 1 (Steps 4 and 5)

- Recall the the following matrices for BV $= \{3,4\}$:

$$B^{-1}N = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}, B^{-1}b = \begin{bmatrix} 12 \\ 12 \end{bmatrix}, \bar{c} = c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} -5 & -4 \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = 0.$$

- We choose x_1 as the **entering variable** (most negative reduced cost).
- We compute the column of the entering variable:

$$B^{-1}A^j = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

- We choose x_3 as the **leaving variable** (minimum ratio test).
- We update BV $= \{3,4\} \cup \{1\} \setminus \{3\} = \{1,4\}$.

Revised Simplex Method: Iteration 2 (Step 3)

- Recall the matrix form:

$$A = \begin{bmatrix} 3 & 2 & 1 & 0 \\ 1 & 6 & 0 & 1 \end{bmatrix}, c = \begin{bmatrix} 5 & 4 & 0 & 0 \end{bmatrix}, b = \begin{bmatrix} 12 \\ 12 \end{bmatrix}.$$

- We compute the following matrices for BV = {1,4}:

$$B = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}, N = \begin{bmatrix} 2 & 1 \\ 6 & 0 \end{bmatrix}, c_{\text{BV}} = \begin{bmatrix} 5 & 0 \end{bmatrix}, c_{\text{NBV}} = \begin{bmatrix} 4 & 0 \end{bmatrix}, B^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{1}{3} & 1 \end{bmatrix},$$

$$B^{-1}N = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{16}{3} & -\frac{1}{3} \end{bmatrix}, B^{-1}b = \begin{bmatrix} 4 \\ 8 \end{bmatrix}, \bar{c} = c_{\text{BV}}B^{-1}N - c_{\text{NBV}} = \begin{bmatrix} -\frac{2}{3} & \frac{5}{3} \end{bmatrix}, c_{\text{BV}}B^{-1}b = 20.$$

- Reduced cost $\bar{c}$ is NOT nonnegative. Current BV is not proven to be optimal.

Revised Simplex Method: Iteration 2 (Steps 4 and 5)

- Recall the the following matrices for BV $= \{1,4\}$:

$$B^{-1}N = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{16}{3} & -\frac{1}{3} \end{bmatrix}, B^{-1}b = \begin{bmatrix} 4 \\ 8 \end{bmatrix}, \bar{c} = c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} -\frac{2}{3} & \frac{5}{3} \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = 20.$$

- We choose x_2 as the **entering variable** (most negative reduced cost).
- We compute the column of the entering variable:

$$B^{-1}A^j = \begin{bmatrix} \frac{2}{3} \\ \frac{16}{3} \end{bmatrix}.$$

- We choose x_4 as the **leaving variable** (minimum ratio test).
- We update BV $= \{1,4\} \cup \{2\} \setminus \{4\} = \{1,2\}$.

Revised Simplex Method: Iteration 3 (Step 3)

- Recall the matrix form:

$$A = \begin{bmatrix} 3 & 2 & 1 & 0 \\ 1 & 6 & 0 & 1 \end{bmatrix}, c = \begin{bmatrix} 5 & 4 & 0 & 0 \end{bmatrix}, b = \begin{bmatrix} 12 \\ 12 \end{bmatrix}.$$

- We compute the following matrices for BV = {1,2}:

$$B = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_{\mathrm{BV}} = \begin{bmatrix} 5 & 4 \end{bmatrix}, c_{\mathrm{NBV}} = \begin{bmatrix} 0 & 0 \end{bmatrix}, B^{-1} = \begin{bmatrix} \frac{3}{8} & -\frac{1}{8} \\ -\frac{1}{16} & \frac{3}{16} \end{bmatrix},$$

$$B^{-1}N = \begin{bmatrix} \frac{3}{8} & -\frac{1}{8} \\ -\frac{1}{16} & \frac{3}{16} \end{bmatrix}, B^{-1}b = \begin{bmatrix} 3 \\ \frac{3}{2} \end{bmatrix}, \bar{c} = c_{\mathrm{BV}}B^{-1}N - c_{\mathrm{NBV}} = \begin{bmatrix} \frac{13}{8} & \frac{1}{8} \end{bmatrix}, c_{\mathrm{BV}}B^{-1}b = 21.$$

- Reduced cost $\bar{c}$ is nonnegative. Hence, current BV is OPTIMAL.
- Optimal **primal** solution: $x_1 = 3,\ x_2 = 3/2$.
- Optimal **dual** solution: $y_1 = 13/8,\ y_2 = 1/8$.

- **Exercise**: Solve the following problem using the revised simplex method. Double check the iterations using the tableau implementation in Python.

$$
\begin{aligned}
\max\ x_0 = 2x_1 + 3x_2 & \\
\text{s.t.} \qquad 2x_1 + \ \ x_2 &\leq 10 \\
x_1 + \ \ x_2 &\leq 6 \\
-x_1 + \ \ x_2 &\leq 4 \\
x_1, x_2 &\geq 0.
\end{aligned}
$$

Simplex Method vs. Revised Simplex Method

- As presented, the revised simplex method is virtually the same as the simplex method.
- However, commercial solvers such as Gurobi always use the **revised** simplex method, for two main reasons:
- The basis-matrix inversion can be implemented efficiently using the previous basis inverse (e.g. the product form of the inverse).
- Elementary row operations in the plain simplex method accumulate round-off errors; the revised method has ways to control these (e.g. occasionally re-inverting the basis matrix from scratch).
- The implementation details of the revised simplex method are advanced topics beyond the scope of this course.

- At each iteration of simplex, we can update B^{-1} using the following:
- Let k be the index of entering variable and r be that of the leavi

$$B_{new}^{-1} = EB_{old}^{-1},$$

where matrix E is an identity matrix except that its rth column is replaced by the vector

$$\eta = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_m \end{bmatrix}, \text{ where}$$

$$\eta_i = \begin{cases} \frac{-a'_{ik}}{a'_{rk}} & if \quad i \neq r \\ \frac{1}{a'_{rk}} & if \quad i = r \end{cases}$$

Revised Simplex Method: Another Example (continued)

$$
\begin{array}{lrl}
\max & 3x_1 + 5x_2 & \\
\text{s.t.} & x_1 & \leq 4 \\
 & 2x_2 & \leq 12 \\
 & 3x_1 + 2x_2 & \leq 18 \\
 & x_1, \quad x_2 & \geq 0
\end{array}
$$

$$
\begin{array}{lrl}
\max & z - 3x_1 - 5x_2 + 0x_3 + 0x_4 + 0x_5 & = 0 \\
\text{s.t.} & x_1 + x_3 & = 4 \\
 & 2x_2 + x_4 & = 12 \\
 & 3x_1 + 2x_2 + x_5 & = 18 \\
 & x_1, x_2, x_3, x_4, x_5 & \geq 0
\end{array}
$$

Revised Simplex Method: Another Example (continued)

$$
\begin{array}{lll}
\max & 3x_1 + 5x_2 & \\
\text{s.t.} & x_1 & \leq 4 \\
 & \quad\;\; 2x_2 & \leq 12 \\
 & 3x_1 + 2x_2 & \leq 18 \\
 & x_1, \quad x_2 & \geq 0
\end{array}
$$

z	x_1	x_2	x_3	x_4	x_5	RHS
1	-3	-5	0	0	0	0
0	1	0	1	0	0	4
0	0	2	0	1	0	12
0	3	2	0	0	1	18

Revised Simplex Method: Another Example (continued)

$$
\begin{array}{lll}
\max & 3x_1 + 5x_2 & \\
\text{s.t.} & x_1 & \leq 4 \\
 & \quad\;\; 2x_2 & \leq 12 \\
 & 3x_1 + 2x_2 & \leq 18 \\
 & x_1, \quad x_2 & \geq 0
\end{array}
$$

z	x_1	x_2	x_3	x_4	x_5	RHS
1	-3	0	0	$\frac{5}{2}$	0	30
0	1	0	1	0	0	4
0	0	1	0	$\frac{1}{2}$	0	6
0	3	0	0	-1	1	6

Revised Simplex Method: Another Example (continued)

$$
\begin{array}{lll}
\max & 3x_1 + 5x_2 & \\
\text{s.t.} & x_1 & \leq 4 \\
 & \quad\;\; 2x_2 & \leq 12 \\
 & 3x_1 + 2x_2 & \leq 18 \\
 & x_1, \quad x_2 & \geq 0
\end{array}
$$

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	0	0	$\frac{3}{2}$	1	36
0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2
0	0	1	0	$\frac{1}{2}$	0	6
0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2